Matthew Stuart Jones

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EDUCATION

University of Kentucky — Projected Graduation, May 2027
Bachelor of Science, Computer Science — *GPA: 4.0/4.0*

- Minors: Mathematics and Vocal Performance
- Artificial Intelligence Certificate
- Study Abroad Experience in Bilbao, Spain (Universidad de Deusto)

RESEARCH EXPERIENCE

NSF Research Experience for Undergraduates (REU), AI for Social Good — Summer 2026
Worcester Polytechnic Institute (WPI) — Worcester, Massachusetts

- Developed ARISES, a novel feature selection method, outperforming all state-of-the-art baselines on 6 of 7 multi-label metrics
- Developed MLX, a counterfactual explainability method using structured gradient descent with label-co-occurrence-weighted gradients, outperforming SHAP and LIME on multi-label, patient-wise explanations for black-box classifiers
- Applied prompt engineering and AI tooling across the pipeline, matching tool to task for efficiency without losing accuracy

Independent Study: Responsible AI Citation & Policy Influence Mapping — Spring 2026
Advisor: Dr. Brent Harrison, University of Kentucky — Human Centered AI, AI Ethics

- Analyzed citation flows between RAI literature (AIES, NeurIPS) and 18 national AI governance frameworks; identified three-way misalignment between paper production (fairness), citation impact (explainability), and policy language (accountability)
- Built end-to-end data pipeline using Semantic Scholar API, intfloat/e5-large-v2 embeddings, and cosine similarity scoring to quantify research-to-policy alignment across a 20,000+ paper corpus
- Presented findings at the 20th Annual University of Kentucky Showcase of Undergraduate Scholars

AI Researcher for ACCESS — October 2024 – Present (incl. Summer 2025)
UK Center for Computational Sciences (CCS) — Lexington, Kentucky

- Built MCP-based semantic router to categorize incoming queries and route them to the appropriate AI agent across the ACCESS cyberinfrastructure ecosystem
- Developed Retrieval-Augmented Generation (RAG) system to assist users in drafting allocation abstracts and proposals based on a corpus of past submissions
- Designed multi-LLM tournament tagger to categorize research proposals by field of science using open-source models deployed on a Grace Hopper cluster via Ollama
- Improved existing AI models through document verification and retraining workflows

INVITED TALKS

Guest Lecturer, HON 251: Technology and the Human Experience (Honors) — Fall 2025
"A Brief Overview of AI Policy" — University of Kentucky

CONFERENCES

- "UK NSF Regional AI Workshop" — Lexington, Kentucky — October 7, 2025
- "UK AI/ML Research Symposium" — Lexington, Kentucky — March 28, 2024; November 12, 2024
- "UK Commonwealth Computational Sciences Summit on AI" — Lexington, Kentucky — October 17, 2023

SKILLS

- **AI/ML Engineering & Explainability:** Built RAG systems, MCP semantic routers, and LLM-based tagging pipelines using LlamaIndex, ChromaDB, sentence-transformers, and Ollama; developed ARISES (feature selection) and MLX (counterfactual explainability via structured, label-co-occurrence-weighted gradient descent), outperforming SHAP and LIME on multi-label, patient-wise explanation tasks; experience with OpenAI API, local open-source models, and prompt engineering.
- **Mathematical Foundations:** Strong background in algorithms, linear algebra, probability, and numerical methods; coursework in sequential decision making, deep learning, and models of computation.
- **Programming & Systems:** Python (NumPy, pandas, scikit-learn) for ML/AI; C and C++; SQL; full-stack web development

(JavaScript, HTML, CSS); Linux; R.

- **Research & Data Pipelines:** Designed end-to-end NLP pipelines for literature analysis at scale — API-based corpus ingestion, embedding generation (e5-large-v2), and cosine similarity scoring across large document corpora; familiar with academic literature APIs (Semantic Scholar, OpenAlex).
- **AI Governance & Policy Communication:** Analyzed citation-flow alignment between RAI research and national AI governance frameworks across 18 countries; translates complex AI and technical concepts for policy, academic, and general audiences; public speaker and technical presenter; fluent in Spanish.

FEATURED COURSEWORK

CS 667	Sequential Decision Making	**CS 463G**	Intro to AI
CS 575	Models of Computation	**CS 315**	Algorithms
CS 485G	Data Ethics	**PPL 306**	Ethics and Civic Leadership
MA 322	Matrix Algebra and Its Applications	**MA 321**	Intro to Numerical Methods
MA 320	Introductory Probability		

HONORS AND AWARDS

- Dean's List — 6 Semesters
- Tau Beta Pi Scholar
- Pigman College of Engineering Marky Campbell Scholarship
- Distinguished Flying Cross Scholarship
- UK Bluegrass Spirit Scholarship
- UK Education Abroad Scholarship

PROFESSIONAL ORGANIZATIONS

Kentucky Alpha Chapter of Tau Beta Pi — Tau Beta Pi Scholar & Chapter Treasurer

- Managed chapter finances for a 120-member engineering honor society, including dues, budgeting, and scholarship requests
- Coordinated food and logistics for chapter meetings and events
- Volunteer judge at Future Cities regional engineering competition

ORGANIZATIONS AND INVOLVEMENT

acoUstiKats A Cappella — Business Manager & Social Media Manager

- Led the group's public presence and built an automated booking pipeline syncing forms with Google Calendar
- Spearheaded the group's first ICCA campaign where I initiated the decision to compete and led competition research
- Created content across platforms, designed and produced group merchandise, and publicized campaign events

UK Men's Chorus — Baritone Section Leader

- Serve as section leader on the group's New York tour, coordinating logistics and welfare for the baritone section
- Part of a leadership team focusing on culture and performance where singers are focused while taking care of each other

PUBLICATIONS

"The Translation Between Responsible AI Research and AI Legislation" — May 2026
20th Annual University of Kentucky Showcase of Undergraduate Scholars

"Predicting Adverse Reactions and Identifying Clinically Meaningful Features in Oncology Patients Using the US FAERS" — *In Progress*
Poster — NSF REU in AI for Social Good, Worcester Polytechnic Institute — Summer 2026
Matthew Jones*, Amruta Soma*, Gandhali Sheth, Dr. Chun-Kit Ngan, Dr. Bo Wang, Dr. Yin Ting Cheung
*Co-First Authors

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REFERENCES

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